2022 10th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)
Amity University, Noida, India. Oct 13-14, 2022

2022 10th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO) | 978-1-6654-7433-7/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICRITO56286.2022.9965087

# Classification of Diabetic Retinopathy Disease with Improved Transfer Learning Techniques using EfficientNets

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***Abstract*—One complication of diabetes is diabetic retinopathy (DR) and is a very common eye disease. It is one of the primary causes of blindness worldwide. Traditionally, a physician identifies the presence of vitals manually when diagnosing the image of the retina of the eye of the people suffering from Diabetic Retinopathy. It is a difficult, time-taking, manual, and skill-oriented task. This paper's aim is to create a system that can automatically analyze the images of the retina and tells about the severity of the disease. This has been achieved by modelling an improved Convolutional Neural Network based model using the transfer learning techniques and various efficientNet models like efficientNet-B1, efficientNet-B2, efficientNet-B3, efficientNet-B4 and efficientNet-B5. This research work uses the dataset provided by Kaggle Community. The efficientNet-B5 model results show an accuracy of 98.91% and a validation kappa score of 0.9308.**

***Keywords— Blindness Detection, Diabetic Retinopathy, Deep Learning, fundus images, Transfer Learning, EfficientNets, Kaggle.***

## I. Introduction

Diabetic Retinopathy is a diabetic condition. It is the primary factor for adult blindness. As per the reports of the World Health Organization, DR is cause behind approximately 5 million blind people, or about 5% of global blindness [1]. Approximately 90% of patients of the DR can be diagnosed in the early phase and complexities can be reduced with good screening and regular check-ups [2]. It damages the vessels in the human eye's. The major issue is that Diabetic Retinopathy is primarily a disease that has no symptoms. It does not show symptoms unless it has progressed to its final stage. The blindness caused by DR is not reversible, however it can be avoided by having regular fundus checks. Manual inspections of the fundus images of retina, on the other hand, are jobs that are difficult and time-consuming. The effect to the blood vessels of retina is shown in fig. 1 [3] where eye with diabetic retinopathy contains leaky blood vessels. The phases of diabetic retinopathy are divided into five categories. [4] as shown in table 1. There are a few microaneurysms in the retina during the stage 1 and stage 2. The stage 3 is characterized by an increase in microaneurysms. Stage 4 leads to blindness [5].

This paper is arranged in such a way that section 2 provides a glimpse of some related works. Section 3 explains the proposed method. Results of the research are given by section 4. Section 5 gives conclusion to the paper.

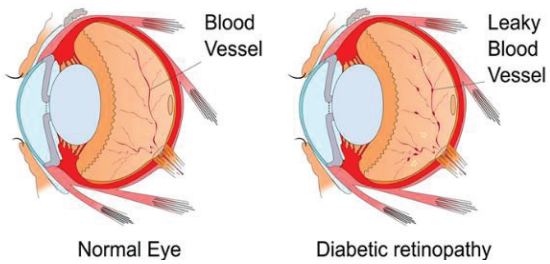


Fig. 1. [5] Diabetic Retinopathy

TABLE I. STAGES OF DIABETIC RETINOPATHY

| Class | Stage of Diabetic Retinopathy |
|---|---|
| Class 0 | No DR |
| Class 1 | Mild DR |
| Class 2 | Moderate DR |
| Class 3 | Severe DR |
| Class 4 | Proliferative DR |

## II. Related Work

Deep learning based Residual Network was used by Grace et al.[5] to classify the DR in its stages. Multi class cross-entropy and Adam optimizer were applied to avoid overfitting. It was hypothesised that the performance of ResNet-50 was significantly better than other models. The model achieved an accuracy of 89.4%.

Karki and Pradnya [6] employed various EfficientNet models to divide DR images into five classes of DR. Intense augmentations were used to improve the model's generalisation. The quadratic kappa score of 0.924377 was achieved by top model using an ensemble model on the APTOS test dataset.

Yash and Snehal [7] used multiclass SVM approach to classify the input images into two types: DR images and normal retinal images. A total of 48 and 90 fundus pictures were used for testing and training. Accuracy of 98.89% was attained. However, accuracy looks very good but it is to be noted that the dataset used is very small.

Yachin [8] developed a model based on deep learning and EfficientNet to classify vision-threatening DR and referable diabetic retinopathy (RDR). Two datasets, APTOS 2019 and EyePACS, were used in the tests. It achieved an AUC of 0.990 for DR and 0.984 for RDR on the EyePACS


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dataset. The APTOS 2019 dataset shows similar results, with AUCs of 0.998 and 0.966 and for DR and referable, respectively.

Rubina et al. [9] compared 13 pre-trained CNNs models based on their accuracy on Kaggle and Messidor dataset. VGG16, Xception, InceptionV3, VGG19, InceptionResNetV2, MobileNet, ResNet50, MobileNetV2, DenseNet121, DenseNet201, DenseNet169, NASNetLarge and NASNetMobile were used on several pre-trained CNN models. They removed the top layer and retrained to find that VGG19, ResNet50, and NASNetLarge were the top three CNN models.

Zago. GT et al. [10] used two CNNs (CNN and pretrained VGG16) to classify images as DR or non-DR based on the presence of patches of the lesion. For training, the database used was DIARETDB1. For testing, the Messidor, DDR, IDRiD, DIARETDB0, Kaggle and Messidor-2 datasets were employed. The Messidor dataset produced the best results having an AUC of 0.912 and a sensitivity of 0.94.

Qi Li et al. [11] created a multi-scale CNN that took advantage of several scales and complimentary visual data. The multi-scale based CNN model outperformed the single-scale based CNN model. On the NJU DR dataset, the MS-CNN model was used. It detected haemorrhages with 95.6% sensitivity and 92.9% accuracy. It also detected micro-aneurysms with 72.7% sensitivity and 88.9% precision.

Alexander S. Shirokanev and Nataly Yu. Ilyasova [12] developed a method for examining the statistical aspects of a preliminary coagulation plan to determine its efficacy in the diagnosis of DR. The distances between points that indicate the plan were investigated. Delaunay triangulation was used to calculate the distances. The distances were used to calculate statistical features. After that, discriminant analysis was used to examine the attributes. The research revealed that updated features can improve the outcome when using KNN or Bayesian classifier.

Morarjee Kolla and T. Venugopal [13] provided Binary Convolutional Neural Networks (BCNN), which considerably boosted the speed of the execution and reduced memory consumption. Their DR classification model was hardware friendly and efficient. It uses dataset from Kaggle. When compared to the standard model, experiments showed the memory computation reduced by 37% and runtime increased by 49%. They compared the performance of their BCNN model to other models like DenseNet, VGG-16, AlexNet, Resnet 50 and Inception V3.

Md. Ahsan et al. [14] demonstrated a model based on the image of the retina and a neural network to predict the DR status. They used the dataset from the Kaggle Community. CNN model was used to represent the VGGnet model. They attained 95.41% accuracy using proposed methodology.

Yefrenes et al. [15] used the VGG fundamental architecture of a CNN. The Messidor database yielded values of 99 % sensitivity, 99.66% accuracy and 98% specificity. The architecture achieved sensitivity, specificity, and accuracy of 98%, 97% and 98.43% respectively, when evaluated with the Kaggle database.

Qiao et al. [16] analyzed the availability of microaneurysms in fundus images utilizing deep learning-based CNN method named PMNPDR. The semantic segmentation technique was used to classify the fundus image.

In some works the dataset used is very small. Large dataset may be used for training and testing purposes. When learning stagnates, no techniques are used to improve the model. Learning rate is kept constant. When performance metric stops improving, learning rate may be reduced, to improve the model. The number of training rounds may also be increased appropriately to improve the performance of the model. Some works classifies the images only into two categories, those are - whether DR is present or not. Not all the five stages of DR are detected. Microaneurysms or exudates are the only features used in the classification. For more advanced processing and more accurate DR picture recognition, other aspects of diabetic retinopathy, such as red lesions, hemorrhages etc., may be used utilizing various biomedical image processing techniques. Due to these gaps some of the model overfits and some have low performance.

## III. The Proposed Method

The most widely utilized methodologies in DR detection are convolutional neural network (CNN) based on learning-based structures and transfer learning [17]. In this paper, the proposed method classifies the images of the retina according to the severity of the disease. It is achieved by developing an improved Convolutional Neural Network model using the transfer learning techniques. Data collection, preprocessing, augmentation, and training as shown in fig. 2 are the various steps followed in the process of classification. This proposed approach first try various efficientNet models like efficientNet-B1, efficientNet-B2, efficientNet-B3, efficientNet-B4 and efficientNet-B5 as done in [6] and tries to improve the performance matric i.e., quadratic kappa score by making following improvements:

- LANCOZ filter is used in image preprocessing step.
- In addition to other techniques employed in [6], images are shifted height-wise and widthwise for image augmentation.
- An additional pooling layer is added in the bottom of the model using GlobalAveragePooling2D() for better feature extraction.
- ReduceLROnPlateau is used to reduce learning rate when learning of the model stagnates.

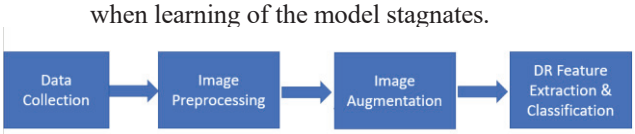


Fig. 2. Various steps in the process of Diabetic Retinopathy Classification

### A. Data Collection

The Kaggle APTOS[18] competition provided the dataset for this proposed work. This dataset contains a vast collection of high-resolution eye's retina images. Medical practitioners categorized the photographs in the dataset from 0 to 4, as indicated in Table I. The dataset has color photographs that vary in dimensions. The image of the retina of the human eye is captured using fundus photography. Figure 3 has some sample images of the retina of the eye of a human being which is labeled with the class of DR at the top of each.


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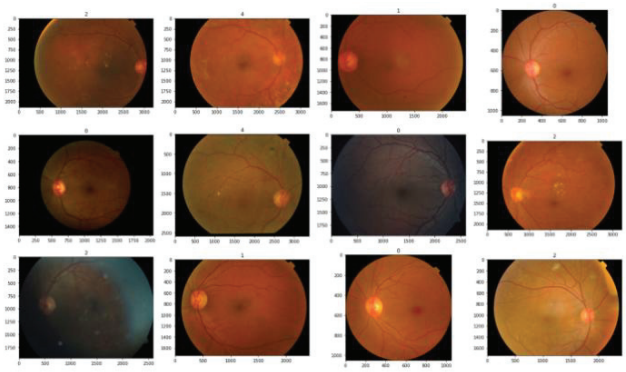

Fig. 3. Sample images in the training set

3662 training images with five levels of severity are present in the dataset. These 5 stages are none, mild, moderate, severe, proliferative stages. Fig. 4 shows the distribution of number of images vs stages of the disease. Fig. 5 shows the top 5 rows in the training data. It has 2 fields: id_code and diagnosis. Id_code is an id to the image. If the image name is abc.png then id_code is abc. Diagnosis represents the class of the DR. It can have 5 values from 0 to 4.

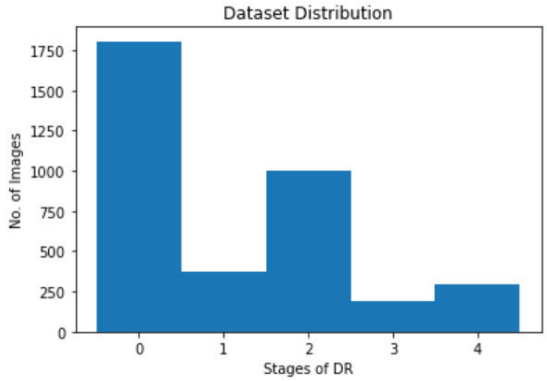


Fig. 4. Number of images of each class

| | id_code | diagnosis |
|---|---|---|
| 0 | 000c1434d8d7 | 2 |
| 1 | 001639a390f0 | 4 |
| 2 | 0024cdab0c1e | 1 |
| 3 | 002c21358ce6 | 0 |
| 4 | 005b95c28852 | 0 |

Fig. 5. Training Data

## B. Image Analysis

The photographs of the retina eye's retina are taken with fundus photography. As may be noticed in the images, in Fig. 3, some of the images are underexposed, overexposed or out of focus. Furthermore, some of the images have dim light and low brightness, it makes it impossible to distinguish between them. To bring out distinguishing features from eye photos, a variety of image preprocessing techniques such as image resizing, and cropping are applied.

## C. Processing of Images

To make images clearer and similar, so that the model can learn features more successfully, images are resized in 256X256 pixels and 300X300 pixels. Filters can be used to remove several types of noise from an image. This approach has used filter with resampling option of Image.LANCOZ. Lancoz is a high-quality down-sampling filter [19]. Figure 6 is a random image from the training image set. It is of size 2300X3500 (approx.) in pixels. After resizing the same image, in Fig. 7, it is 256X256 in pixels.

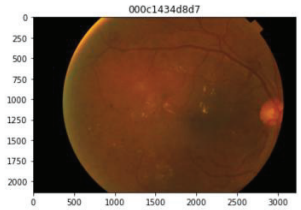


Fig. 6. Images before resizing

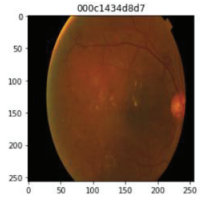


Fig. 7. Image after resizing

## D. Image Augmentation

This is one of the most used procedures for generating data robustness by making some extra images of the dataset[20]. Augmentation is achieved using rotation techniques, flips the images horizontally and vertically, cropping the images, padding the images by filling the points outside the input boundaries and shifting the image height-wise and widthwise. It has been implemented using the ImageDataGenerator. Figure 8 shows the different augmented images generated from one image only.

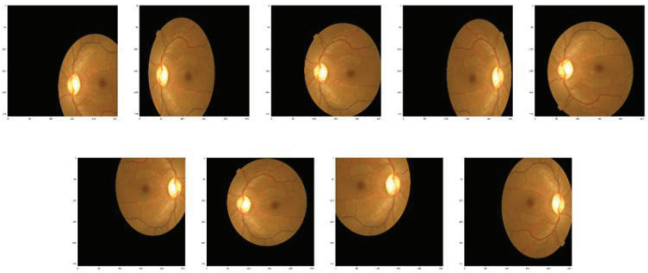

Fig. 8. Augmentations of a image

## E. DR Feature Extraction and Classification

EfficientNets(B1-B5) are the backbone of this research. Google AI introduces EfficientNet [21], which can increase the performance of DR severity classification. It is proposed in this study to employ pretrained models for better and faster convergence. These pre-trained models were learnt using the ImageNet dataset [20]. They are imported using Keras built-in libraries in python programming language [20]. For initialization of weights, it used the weights of the trained-on ImageNet. Than some more layers are added in the bottom model like pooling layer using GlobalAveragePooling2D(), dropout and dense. One epoch is entire forward and backward passes through the network. As it has used the pre-trained models, epoch was set at 20. The model of an epoch has been saved only if kappa score has improved. Adam optimizer [11][22] for all the models with an initial learning rate of 0.00005. Early stopping is used. It is often beneficial for the models to slow down the learning rate by a factor of 2-10 when learning becomes stagnant [23]. When metric stops improving, ReduceLROnPlateau is used to Reduce learning rate. The images of the retina of the eye are classified into 5 categories based on the severity of DR using various EfficientNet models from efficientNet-B1 to efficientNet-B5.

# IV. Results

The proposed method has been implemented on Windows 10 Enterprise x64 based PC having Intel(R) Xeon(R) W-2133 CPU. Processor frequency was 3.60GHz,


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3600 Mhz. Processor has 6 Cores, 12 Logical Processors having 64 GB RAM. Python 3 is used to implement the models. Performance metric used in this work is kappa score. The configurations of the proposed work have improved the kappa score of the classification in comparison to [6]. EfficientNet-B5 model with size of image as 300X300 performs the best among all the models. It has outperformed all other models with a kappa score of 0.9308 achieved an accuracy of 0.9891 with a loss value of 0.0885. In addition to kappa score, loss and accuracy are also measured and described in table II. Variations of kappa score on each epoch for each model is shown in fig. 9.

TABLE II. RESULTS FOR EACH MODEL

| Model (Input Size) | Kappa Score [6] | Kappa Score [proposed work] | Loss [proposed work] | Accuracy [proposed work] |
|---|---|---|---|---|
| EfficientNet-B1 (256x256) | 0.9061 | 0.9261 | 0.0934 | 0.9782 |
| EfficientNet-B2 (256x256) | 0.9060 | 0.9267 | 0.0929 | 0.9891 |
| EfficientNet-B3 (256x256) | 0.9174 | 0.9198 | 0.1039 | 0.9618 |
| EfficientNet-B4 (256x256) | overfitted | 0.9182 | 0.0999 | 0.9764 |
| EfficientNet-B5 (256x256) | 0.9078 | 0.9276 | 0.0916 | 1.0000 |
| EfficientNet-B5 (300x300) | 0.9040 | 0.9308 | 0.0885 | 0,9891 |

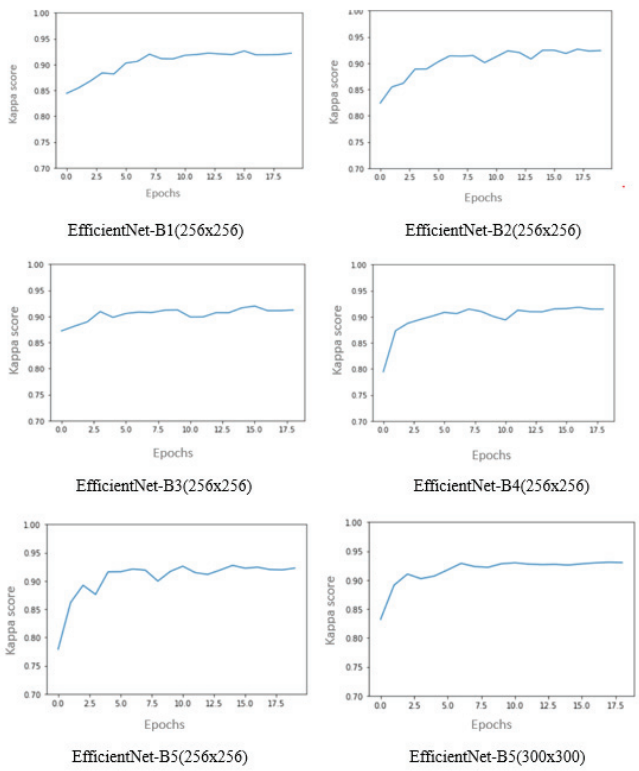


Fig. 9. Kappa score variations as per epoch for each model

As the epoch increases kappa score improves gradually. When kappa score stopped improving, we reduced the learning rate, and it helped in improving kappa score future. However, it is to be noted that only reducing learning rate 2-3 times helped, after that reducing it more didn't help in improving kappa score, hence early stopping was used for EfficientNet-B3, EfficientNet-B4 and EfficientNet-B5 (300x300) to save future computation. EfficientNet-B1 achieved its highest kappa score of 0.9261 at 16/20 epoch. EfficientNet-B2 achieved its global maxima of 0.9267 at 18/20 epoch. EfficientNet-B3 has its peak 0.9198 at 16/20 epoch. EfficientBet-B4 has its maximum value 0.9182 at 17/20. EfficientNet-B5 achieved its highest kappa score 0.9276 and 0.9308 at 15/20 epoch and 18/20 for image size 256x256 and 300x300 respectively. As the epoch increases, accuracy increases and loss decreases. High accuracy and low loss are better for any model.

## V. CONCLUSION

Automated systems for DR detection are important for identifying DR at a preliminary phase. In this research work, an improved method is presented to classify the images of the retina of the eye into different severity levels of the DR. Transfer learning powered by efficientNets models is implemented to detect DR in early phases. Performance matric i.e., kappa score of various efficient algorithms is calculated and based on the results EfficientNet-B5 is suggested to be used.

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